NYSC Registration Dataset — SQL Practice Exercises

Each question is designed for progressive SQL skill development. Answers are intentionally hidden for self-practice.

SECTION 1 — Basic SELECT & Filtering

1. Retrieve all columns from the table.

2. Show only the Fullname, Age, and Gmail of all registrants.

3. List all registrants whose age is greater than 25.

4. Display all records where the origin state is 'Lagos'.

5. Find all registrants who graduated with 'First Class'.

6. Retrieve the names of registrants whose preferred state is 'Abuja' and origin state is 'Kano'.

7. Show all registrants whose email address ends with '@gmail.com'.

SECTION 2 — Sorting & Distinct

1. List all unique origin states in the dataset.
2. Display all registrants ordered by Age in descending order.
3. Show the top 5 oldest registrants.
4. Find all unique institutions in the dataset.

SECTION 3 — Aggregation & Grouping

1. Count the total number of registrants.

2. Find the average age of all registrants.

3. Count how many registrants come from each origin state.

4. Show the number of registrants from each preferred state, ordered from most to least preferred.

5. Display the number of graduates by each graduating honors category.

6. Find the average age of registrants for each institution.

SECTION 4 — Conditional & Filtering Aggregates

1. Find all institutions that have more than 100 registrants.
2. Count how many registrants from Rivers State prefer to serve in Lagos.
3. Calculate the average age of registrants who graduated with Second Class Upper.
4. List all courses that have more than 50 graduates.
5. Find the youngest and oldest registrant.

SECTION 5 — Subqueries & Analytical Thinking

1. Find registrants whose age is above the overall average age.

2. List all registrants who attended an institution located in the same state as their origin state.

3. Identify the most common course of study among registrants.

4. Find the preferred state of the youngest registrant.

5. Show the institution(s) that produced the highest number of First Class graduates.

6. List all origin states whose registrants prefer to serve outside their home state.

7. Determine which institution state contributes the most registrants.

8. Find all registrants whose course study matches any course studied at University of Lagos (using a subquery).

BONUS CHALLENGES

1. Create a query to find the percentage distribution of each graduating honor category.

2. Find the top 3 most common institutions.

3. Show for each origin state: total registrants, average age, and most common preferred state.

4. Write a query that finds the difference between the average ages of male and female registrants (if a gender column existed — describe how you'd modify the dataset).

5. Using window functions (if supported), rank registrants by age within each institution.